

Extracting the Affective Content of Fidgeting in Adults with ADHD via Machine Learning and a Hand-held Soft Tangible Device

Nahid Nasiri
nnasiri@ucsc.edu
University of California, Santa Cruz
Santa Cruz, California, USA

Katherine Isbister
kisbiste@ucsc.edu
University of California, Santa Cruz
Santa Cruz, California, USA

Julie B Schweitzer
jschweitzer@ucdavis.edu
Department of Psychiatry and
Behavioral Sciences and MIND
Institute
UC Davis, USA

Jared Borden
jborden@ucdavis.edu
Department of Psychiatry and
Behavioral Sciences and MIND
Institute
UC Davis, California, USA

Peter S Cottrell
pcottrel@ucsc.edu
University of California, Santa Cruz
Santa Cruz, California, USA

Daniel Shapiro
dgs@ucsc.com
Daniel Shapiro
Santa Cruz, California, USA

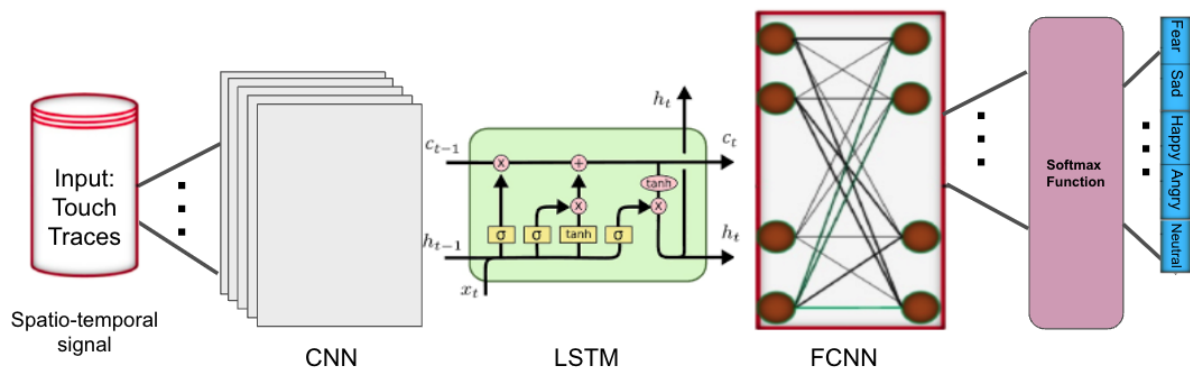


Figure 1: Architecture to Classify Emotion Inducing Video Stimulus from Spatio-temporal Touch Traces

ABSTRACT

This paper examines the task of extracting affect from the fidgeting behavior of adults with Attention Deficit/Hyperactivity Disorder (ADHD) engaged in emotional self-regulation experiments. We describe a fidgeting appliance developed to capture a tactile signal from spontaneous hand fidgeting, clarify its affordances, and present machine learning methods for processing this data. We show that hand-fidgeting carries a strong affective signal by recognizing the affective state produced by emotion-inducing videos and predicting self-assessed measures of anxiety and anxiety management collected after a standardized stress test. We conclude by discussing the potential impact of practical methods for data collection and analysis of fidgeting behavior on ADHD detection and management.

KEYWORDS

Affective content, touch, fidgeting behaviors, ADHD, emotional self-regulation, tactile signals, machine learning

ACM Reference Format:

Nahid Nasiri, Katherine Isbister, Julie B Schweitzer, Jared Borden, Peter S Cottrell, and Daniel Shapiro. 2024. Extracting the Affective Content of Fidgeting in Adults with ADHD via Machine Learning and a Hand-held Soft Tangible Device. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3650856>

1 INTRODUCTION

Determining the affective content of human input is a key subtask of affective computing [42], as it is the first step of a processing loop aimed at enabling human-computer interaction in more visceral terms. Affect analysis has been pursued from image data, audio, and text, for example, via facial expression recognition [30], stress determination from speech [2], emotion from prosody [3], sentiment

analysis from social media [5], these in combination with physiological measures [23], and via multi-modal analyses [34]. In contrast, we focus on an under-explored channel: affect analysis from tactile signals generated by fidgeting behavior. This is particularly pertinent for individuals with ADHD, who commonly experience excess motor activity [27], including fidgeting and challenges in affective control.

In this context, we develop a novel device: a hand-held, sensor-enabled, soft-tangible 'fidget ball' [13] to capture the nuanced affective signals inherent in hand-fidgeting. We conduct a study that engages individuals with ADHD in tasks concerning affective self-regulation, and we employ machine learning techniques to analyze the tactile data obtained. Our approach demonstrates the feasibility of extracting affective content from fidgeting patterns. We establish that hand-fidgeting carries a nuanced affective signal that can be harnessed to predict emotional states induced by external stimuli and assess the efficacy of affective self-regulation strategies. These findings suggest a pathway for the development of unobtrusive, intelligent appliances designed to support self-regulation for individuals with ADHD.

We begin by establishing the contextual framework. Section 2 describes related work on devices, fidgeting, and affect analysis. Section 3 discusses the fidget ball we developed for capturing affective touch, its sensor suite, and affordances. Section 4 documents the human subject experiments we performed, while Section 5 presents the machine learning (ML) architecture and methods we employed to analyze the associated fidgeting traces. We present and discuss experimental results in Sections 6 and 7, followed by limitations of the study in Section 8. We present conclusions and future work in Section 9.

2 RELATED WORK

We separate our discussion of related work into three sections corresponding to the areas of study bridged by this research: fidgeting devices, fidgeting behavior in ADHD, and the technology of affect analysis.

2.1 Fidget Devices

In HCI research, studies by Da et al. [9] and Karlesky and Isbister [25] highlight the use of fidget items in enhancing concentration and managing mood. The development of early 'fidget widgets' by Karlesky and Isbister [24] integrated tactile and digital interactions, marking a significant step in exploring physical interaction in digital interfaces. Advances in technology like Skweezee [41] and CuddleBits [6] have improved the capture of tactile dynamics, influencing the design of smart fidget devices for affective state and self-management research. Skweezee improved smart fidget devices with precise pressure detection and tactile feedback, while CuddleBits offered lifelike touch interactions using soft robotics, advancing tactile dynamics in smart fidget device research.

2.2 Fidgeting in ADHD

Fidgeting has long been recognized as a key symptom of ADHD. In the "optimal stimulation theory of ADHD" [44, 46], persons with ADHD seek out stimulation in response to conditions of over-arousal (e.g., heightened stress) or under-arousal (i.e., boredom) in

an attempt at homeostasis. Providing high sensory and motor stimulation within tasks improves cognitive and academic performance for the ADHD group and "normalizes" their behavior [45, 46]. Adolescents with ADHD exhibited better cognitive performance on an attention task given more intense fidgeting [18], supporting theories [27, 35, 37] that hyperactivity may serve a "purposeful function" to enhance alertness and regulation. While overt hyperactivity is mainly associated with childhood, subtler, finer-grained frequent movements (e.g., leg movements, doodling, clicking objects, tapping) are highly common in adult ADHD. Both children and adults with ADHD have challenges with emotional regulation and the ability to moderate their stress response [11].

While fidgeting plays an important role in ADHD, little is known about the characteristics of fidgeting in adulthood or whether its content, especially as captured via an external device, can be used to gain insight into ADHD behaviors. Our work tests how a fidget device and natural fidgeting might reveal emotional states and coping in adults with ADHD.

2.3 Affect Analysis

Research on affect analysis has a long history, connecting the development of psychological theories of affect [39] to technology for extracting affective content across diverse data modalities [42]. Early efforts included finding physiological signatures of affective states, such as minute muscle movements associated with touch using a sentograph [7]. Subsequent work explored automatic affect analysis from physiological measures like EEG and ECG [23]. Over time, the focus shifted from knowledge-based techniques with hand-engineered features [47] to more recent learning mechanisms and hybrid approaches [42]. Technology for affect analysis has an obvious role in assistive contexts [1], for example, to improve machine interaction with autistic individuals by employing physiological measures to infer emotional state via a machine learning model [8], analogous to our work on fidgeting data here.

The domain of affect recognition from touch, although comparatively underexplored, is gaining significance. For example, keyboard strokes and touchpad patterns have been shown to indicate emotional states using features like typing speed and force patterns over time [10, 21, 29, 32, 40]. Horii et al. utilize probabilistic neural networks and pressure data to infer six emotional states but with simulated data [22], showcasing the potential of integrating diverse data sources for emotion analysis. Work by Ghosh et al. in using typing pressure and speed on mobile devices for emotion classification [14, 15], along with Shapiro et al.'s deep learning-based automation of feature extraction [38], underscores the potential of such technologies in user-centric applications. Our research utilizes deep learning to analyze emotions through fidgeting behaviors, introducing a novel modality for assistive computing but with the same objective; to improve user interaction through better alignment with their emotional state.

3 CAPTURING A FIDGETING SIGNAL: THE FIDGET BALL

We have developed a smart fidget device to capture the details of fidgeting behavior. Unlike commercial fidget toys designed to absorb attention, this device aims to provide appropriate sensory

stimulation without causing distractions, making it suitable for laboratory and field use by a broad adult demographic. The design is informed by prior surveys [24] and inspired by common stress-relief objects. It employs a spherical form factor to resemble items like stress and tennis balls and supports both one- and two-handed use. It affords common gestures like squeezing and stroking and supplies tactile feedback. The materials were selected for desirable qualities in fidget objects, like pliability and softness.



(a) Core hardware components used in the first iteration of the ball, displayed from left to right: Roller toy, microcontroller, MicroUSB cable, and six 18.5 mm force-sensitive resistors

The engineering design focused on balancing rigidity for electronic protection and tactile qualities for user interaction. Also, the design is intentionally neutral to present a wide variety of tactile affordances while minimizing bias in the elicitation of affective touch. A flexible rubber ball, about 4 inches in diameter, forms the device's base, housing an Adafruit Feather M0 microcontroller and RTC daughterboard. The initial model [38] only used pressure sensors for data collection. However, user feedback led to adding an inertial measurement unit to track more diverse interactions, like rolling or tossing. The device is tethered for power and data transmission. Its fabric cover, neutral in color, conceals sensor locations while allowing easy identification of buttons and is chosen to minimize visual distraction and potential color-based biases. Future iterations might offer customization options for different user preferences, like playful designs for children.

4 ELICITING FIDGETING BEHAVIOR VIA PSYCHOLOGICAL EXPERIMENTS

We conducted several experiments, approved by the university Institutional Review Board (IRB), to elicit fidgeting behavior in emotionally demanding contexts. Participants were between 18

- 50 years of age. They screened to ensure a childhood history of ADHD, as well as current significant challenges with ADHD without evidence of current major psychiatric distress (e.g., major depression, suicidality, psychosis). Licensed Ph.D. psychologists made all diagnostic determinations and approvals for participation in the experiments.

Participants were tested in a research laboratory at a university center for neurodevelopmental disorders. They were instructed to hold a fidget device, and a researcher demonstrated various ways to interact with it. They sat facing a computer with the device tethered by a cord. During most sessions, the participant was alone in the room, observed by a researcher from an adjoining area.

Emotion Induction: The analysis in this paper focuses on the relationship between fidgeting and emotional states. Conditions tested included the presentation of validated film clips known to elicit specific emotions [17], each 2 minutes long, in a fixed order: anger, sadness, fear, and amusement. We interspersed each clip with a neutral nature scene of the same length to reduce carry-over effects and provide a brief recovery period. Data were collected from spontaneous fidgeting in response to these videos for later analysis [12, 48].

Social Stress: We employed the Trier Social Stress Test (TSST) to elicit fidgeting under conditions of cognitive and emotional load. The TSST is known to induce significantly greater self-reported stress in adults with ADHD compared to controls [20]. TSST participants have 5 minutes to prepare a speech explaining why they are the ideal applicant for their dream job and then deliver this speech over a video link to a remote evaluation panel. Following the TSST, participants were informed they did "a good job" and given a 5-minute recovery period. They then rated their social anxiety during the TSST using the State Social Anxiety (SSA) Questionnaire [26, 36] and the State Emotion Regulation Questionnaire (SERQ) [16, 26], both standardized instruments for evaluating affective state and efforts at self-regulation. Participants were debriefed post-session and informed that their performance was not evaluated. Refer to Appendix A for the complete set of questions in the SSA and SERQ.

5 EXTRACTING AFFECT FROM TOUCH VIA MACHINE LEARNING

We address the problem of extracting affect from touch by employing end-to-end ML techniques, which are particularly suited to this task due to the opaque nature of touch signals; it is difficult to identify a set of hand-engineered features that carry affective information or to define an algorithm that reliably combines such features to determine affect. In contrast, ML methods learn a mapping from touch to affect given sufficient data. The sequential nature of touch makes it especially amenable to ML architectures designed for interpreting time series data in classification or regression tasks [31].

5.1 Data

Our research data consists of touch traces obtained from 39 participants instructed to "hold the fidget ball" during the Emotion Induction and TSST tasks. We focused solely on pressure sensor data, thresholded to remove noise, and segmented into continuous

sequences separated by pauses of at least half a second. Whitespace beyond 1/2 second was removed. This process yielded 210 minutes of active touch. We segmented the fidgeting sequences into training, validation, and testing sets, with proportions of 60%, 20%, and 20%, respectively. Each partition contained data from all participants. This dataset was ample for all learning tasks.

5.2 Machine Learning Model

We employ a deep learning model to associate the tactile content of fidgeting signals with experimental metrics via supervised learning. Our architecture, shown in Figure 1, inputs half-second windows of touch data. It employs Convolutional Neural Network layers for feature extraction, Long Short-Term Memory layers for capturing dependencies across time windows, and fully connected neural network layers to integrate the inferred features. Depending on the task, the output is generated either through a softmax layer for classification or an additional FCNN layer for regression. The architecture employs 1D CNN across data from different sensors within each half-second window, complemented by the attention mechanism, to facilitate an enhanced spatial analysis of data collected during participant interactions. We train the model using labeled sequences of half-second data windows, with multiple overlapping sequences shifted by a single data sample. All sequences are presented in a fixed but randomized order during training. The model achieved its best performance with an optimal setup that included 6 CNN layers, 1 LSTM layer, 2 FCNN layers, a learning rate set to 0.01, a batch size of 64, and a training duration of 80 epochs.

6 EXPERIMENTAL RESULTS

We present two distinct ML analyses that examine the affective content of the fidgeting signal. The first recognizes the emotional states elicited by video stimuli in the emotion induction experiment. The second predicts participants' self-assessed measures of anxiety and anxiety management obtained after the TSST test.

6.1 Recognizing emotional state from a fidgeting trace

We framed the recognition of affective states produced by emotion-inducing videos as a 5-way classification task, from sequences of half-second windows of touch to a probability vector over [angry, happy, sad, neutral, fear]. Reading down the diagonal, the model achieved accuracies of 68.1% for 'Angry,' 49.8% for 'Sad,' 54.2% for 'Fear,' 68.4% for 'Happy,' and 48.0% for 'Neutral.' These results significantly surpass the 20% baseline of random choice. Interestingly, when the model misclassifies a state, it predicts fear and angry next, in roughly equal measure. This suggests that one common set of confounds, closely correlated with both fear and angry, appears in the fidgeting signal for all states.

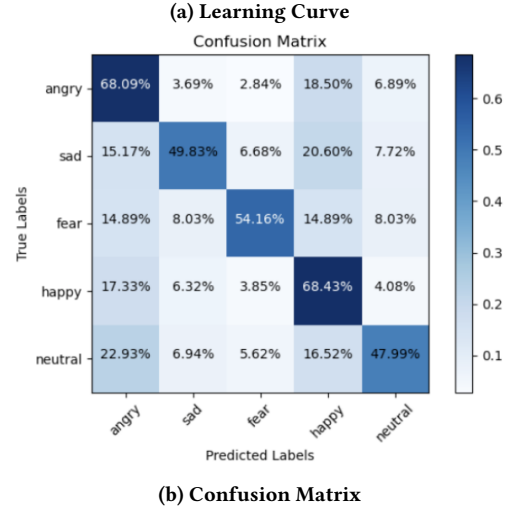
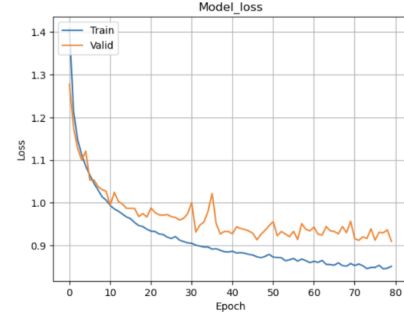


Figure 3: Classification results for emotion-inducing video experiment

6.2 Predicting Self-Assessed Anxiety Scores from Fidgeting Behavior

Utilizing machine learning models, we analyzed self-reported measures of affect obtained following the TSST test. Our method predicted the specific numerical values of participants' responses, with precision rates varying from 52.32% to 81.11%, as detailed in Table 1.

Participants filled out two distinct questionnaires: the SSA gauges the level of social anxiety through questions like, "I worried about what other people thought of me," with answers on a 5-point Likert scale. The SERQ assesses self-regulation efforts via questions like "I kept my emotions to myself" on a 7-point scale. See Table 1 in Appendix A for the full list of questions and answer distributions.

6.2.1 SSA Analyses: We report the Mean Absolute Error (MAE) for each question in the SSA, comparing the model's predictions against the actual response probabilities. Figure 4a shows that the model substantially surpasses (i.e., has much lower MAE values than) the MAE baseline calculated from the median guess. We also performed a classification analysis, treating each participant's numerical response as a categorical variable, and obtained accuracies ranging from 48% to 68.4%, substantially exceeding the 20% threshold associated with random choice.

6.2.2 SERQ Analyses: Figure 4b reports results from a second ML model trained to predict answers for the eight questions on the SERQ. As before, we compare MAE values for the prediction against a baseline representing the median guess. Notably, all MAE values are slightly elevated compared to those of the SSA, which is consistent with the wider scale of the SERQ. The model’s error is significantly less than the baseline, indicating that the fidgeting signal contains information about affective self-regulation measured by the SERQ.

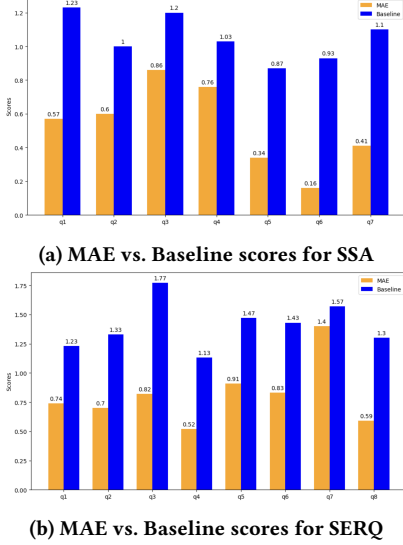
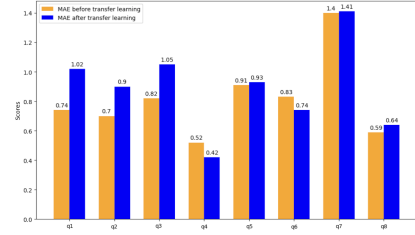


Figure 4: SSA and SERQ Performance Analysis

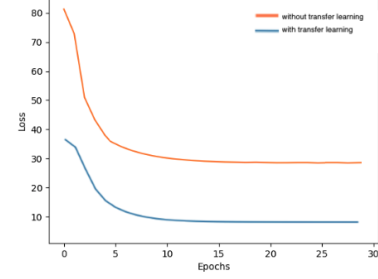
6.2.3 Transfer Learning Results. Given that three ML models with similar structures succeeded in three affect analysis tasks, we investigated the hypothesis that the architectures leveraged a common set of features by conducting a transfer learning study. We altered the final layer of the trained SSA model to output a 7-point scale and retrained the entire model to predict SERQ answers from this initialization. Figure 5a shows that the transfer model produced nearly identical performance as the SERQ-specific model, indicating that a single set of touch features carries information about both anxiety levels and efforts at regulating anxiety. The learning curve in 5b shows the desired benefit of successful transfer; lower initial loss, quicker arrival at lower loss levels and a lower asymptote.

7 DISCUSSION

Our study investigated the affective content of hand-fidgeting in adults with ADHD. Our design for the fidget ball affords affective touch sufficient for subtle tactile analyses, while our experiments show that the fidgeting signal is strong; it supports 50-70% accuracy in predicting five emotional states from touch, and Mean Absolute Error values far below chance in predicting self-assessed measures of anxiety and anxiety management. Our ability to predict these measures suggests that fidgeting serves both expressive and aspirational roles in individuals with ADHD, contributing to the understanding of its function. Our transfer learning work indicates



(a) MAE Results for SERQ With and Without TL



(b) SERQ Outcomes with and without TL

Figure 5: Transfer Learning Performance Analysis

that a common base of touch features carries several types of affective content, suggesting the potential to broaden our analysis of affect from fidgeting in ADHD.

8 LIMITATIONS

This paper reports on an initial investigation into the affective content of fidgeting, which has several limitations. First, prior research makes it clear that people who fidget with objects are attracted to a wide range of affordances that may not be represented in this first device (see [9] and [25]). This would impact the character of the tactile signal and the scope of our conclusions but not our key result, which is that fidgeting carries several types of affective content.

Second, the ML analyses could have obtained their results through unexpected pathways, calling our interpretations into doubt. For example, because our training, test, and validation sets include data from all participants, the ML model for the SSA and SERQ analyses could, in principle, learn to identify individuals from their fidgeting signal and essentially *recall* their self-assessed anxiety scores from the supervised data. However, this memorization path is only available in the affect recognition task (section 6.1), where the learned model demonstrably responds to variations in touch within and across participants induced by the video stimulus. Recognizing individuals by their fidgeting behavior would also be a significant result.

A third potential limitation is that the self-assessment data collected by the SSA and SERQ questionnaires is weak or biased, as the answers require introspection, and ADHD is often associated with reduced introspection. However, a review showed that the ratings vary by item in an expected way given the task, indicating that the participants provided valid responses. Our conclusion that

fidgiting in ADHD carries information about affective state and affect regulation follows from this data.

Lastly, while the accuracy of our analyses might be enhanced via several techniques, our initial study aimed to demonstrate an effect: that fidgiting carries affect in ADHD.

9 CONCLUSIONS AND FUTURE WORK

While we have known that functional touch carries affect [15], and that touch can purposefully communicate affect [19], this work shows that unconstrained fidgiting expresses affect as well. Given fidgiting carries affect, it may contain other implicit features; we are currently examining the conjecture that fidgiting communicates elements of the cognitive state [43] by using ML models to predict task performance in conditions of cognitive load (with respect to processing speed, response inhibition, and cognitive interference). If successful, the work will relate the tactile content of fidgiting to established measures of cognitive function.

Given that affective self-regulation impacts fidgiting, the reverse may also be true; altering fidgiting behavior might impact affect regulation. (The bi-directional effect is illustrated by the classic pencil-between-the-teeth experiment [28] where mood improves by forcing a smile). This would be especially important for individuals who experience challenges with emotional regulation, such as in ADHD. We are pursuing the exciting possibility of constructing a smart appliance that would support such individuals by (a) visualizing their affective state and efforts at affective self-regulation, (b) predicting the evolution of their affective state, and (c) offering suggestions that might help achieve desirable outcomes. The last requires a language for describing fidgiting interventions and a demonstration that altering physical expression alters affect and cognition (analogous to the premise of dance therapy). We are taking the first steps by developing ML models that recognize named gestures in fidgiting traces and seek to relate gesture sequences to affective outcomes in experiments like those reported here.

Lastly, there is reason to believe that fidgiting in ADHD is distinct from fidgiting in the general populace because of the increased role of movement in ADHD [4, 27]. If this conjecture proves true, an ML analysis of a fidgiting signal could improve our ability to diagnose ADHD, primarily based on qualitative measures in current practice [33].

This work was conducted as part of a randomized controlled trial with full consent of study participants and data anonymization. Applying our work beyond this protected context requires care. For example, data privacy becomes a concern if instrumented fidget objects can be used to detect ADHD while providing a visible therapeutic device, which requires ethical analysis. We will bring an ethical lens to the future development of this technology and approach.

ACKNOWLEDGMENTS

The National Institutes of Mental Health supported this work. We thank the clinical research team and the study participants for their contribution.

REFERENCES

- [1] Omar AlZoubi, M Sazzad Hussain, and Rafael A Calvo. 2014. Affect-aware assistive technologies. In *Assistive Technology for Cognition*. Psychology Press, 47–64.
- [2] Alice Baird, Andreas Triantafyllopoulos, Sandra Zänkert, Sandra Ottl, Lukas Christ, Lukas Stappen, Julian Konzok, Sarah Sturmbauer, Eva-Maria Meßner, Brigitte M. Kudiella, Nicolas Rohleder, Harald Baumeister, and Björn W. Schuller. 2021. An Evaluation of Speech-Based Recognition of Emotional and Physiological Markers of Stress. *Frontiers in Computer Science* 3 (2021). <https://www.frontiersin.org/articles/10.3389/fcomp.2021.750284>
- [3] Tanja Bänziger, Didier Grandjean, and Klaus R Scherer. 2009. Emotion recognition from expressions in face, voice, and body: the Multimodal Emotion Recognition Test (MERT). *Emotion* 9, 5 (2009), 691.
- [4] Russell A Barkley et al. 2006. Attention-deficit hyperactivity disorder: A handbook for diagnosis and treatment. (2006).
- [5] Sérgio Barreto, Ricardo Moura, Jonnathan Carvalho, Aline Paes, and Alexandre Plastino. 2023. Sentiment analysis in tweets: an assessment study from classical to modern word representation models. *Data Mining and Knowledge Discovery* 37, 1 (Jan. 2023), 318–380. <https://doi.org/10.1007/s10618-022-00853-0>
- [6] Laura Cang, Paul Bucci, and Karon E MacLean. 2015. Cuddlebits: friendly, low-cost furballs that respond to touch. In *Proceedings of the 2015 ACM on International Conference on Multimodal Interaction*. ACM, 365–366.
- [7] Manfred Clynes. 1978. *Sentics: The Touch of the Emotions*. Anchor Press/Doubleday.
- [8] Karla Conn, Changchun Liu, Nilanjan Sarkar, Wendy Stone, and Zachary Warren. 2008. Affect-sensitive assistive intervention technologies for children with autism: An individual-specific approach. In *RO-MAN 2008-The 17th IEEE International Symposium on Robot and Human Interactive Communication*. IEEE, 442–447.
- [9] Suzanne B da Câmara, Rakshit Agrawal, and Katherine Isbister. 2018. Identifying Children's Fidget Object Preferences: Toward Exploring the Impacts of Fidgeting and Fidget-Friendly Tangibles. In *Proceedings of the 2018 Designing Interactive Systems Conference*. ACM, 301–311.
- [10] Clayton Epp, Michael Lippold, and Regan L. Mandryk. 2011. Identifying emotional states using keystroke dynamics. In *Proceedings of the 2011 annual conference on Human factors in computing systems - CHI '11*. ACM Press, Vancouver, BC, Canada, 715. <https://doi.org/10.1145/1978942.1979046>
- [11] Stephen V Faraone, Tobias Banaschewski, David Coghill, Yi Zheng, Joseph Biederman, Mark A Bellgrove, Jeffrey H Newcorn, Martin Gignac, Nouf M Al Saud, Iris Manor, et al. 2021. The world federation of ADHD international consensus statement: 208 evidence-based conclusions about the disorder. *Neuroscience & Biobehavioral Reviews* 128 (2021), 789–818.
- [12] Cristina Fernández, Juan C Pascual, Joaquim Soler, Matilde Elices, Maria J Portella, and Enrique Fernández-Abascal. 2012. Physiological responses induced by emotion-eliciting films. *Applied psychophysiology and biofeedback* 37 (2012), 73–79.
- [13] Anna Flagg and Karon MacLean. 2013. Affective Touch Gesture Recognition for a Furry Zoomorphic Machine. In *Proceedings of the 7th International Conference on Tangible, Embedded and Embodied Interaction (TEI '13)*. ACM, New York, NY, USA, 25–32. <https://doi.org/10.1145/2460625.2460629> event-place: Barcelona, Spain.
- [14] Surjya Ghosh, Niloy Ganguly, Bivas Mitra, and Pradipta De. 2017. *TapSense: combining self-report patterns and typing characteristics for smartphone based emotion detection*. <https://doi.org/10.1145/3098279.3098564>
- [15] Surjya Ghosh, Kaustubh Hiware, Niloy Ganguly, Bivas Mitra, and Pradipta De. 2019. Emotion detection from touch interactions during text entry on smartphones. *International Journal of Human-Computer Studies* 130 (Oct. 2019), 47–57. <https://doi.org/10.1016/j.ijhcs.2019.04.005>
- [16] James J Gross and Oliver P John. 2003. Individual differences in two emotion regulation processes: implications for affect, relationships, and well-being. *Journal of personality and social psychology* 85, 2 (2003), 348.
- [17] James J Gross and Robert W Levenson. 1995. Emotion elicitation using films. *Cognition & emotion* 9, 1 (1995), 87–108.
- [18] TA Hartanto, CE Krafft, Ana-Maria Iosif, and Julie B Schweitzer. 2016. A trial-by-trial analysis reveals more intense physical activity is associated with better cognitive control performance in attention-deficit/hyperactivity disorder. *Child Neuropsychology* 22, 5 (2016), 618–626.
- [19] Alicia Heraz and Manfred Clynes. 2018. Recognition of Emotions Conveyed by Touch Through Force-Sensitive Screens: Observational Study of Humans and Machine Learning Techniques. *JMIR Mental Health* 5, 3 (Aug. 2018), e10104. <https://doi.org/10.2196/10104>
- [20] Daniel F Hermens, Leanne M Williams, Ilario Lazzaro, Stephanie Whitmont, Dmitriy Melkonian, and Evian Gordon. 2004. Sex differences in adult ADHD: a double dissociation in brain activity and autonomic arousal. *Biological psychology* 66, 3 (2004), 221–233.
- [21] Javier Hernandez, Pablo Paredes, Asta Roseway, and Mary Czerwinski. 2014. Under Pressure: Sensing Stress of Computer Users. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. ACM, New York, NY, USA, 51–60. <https://doi.org/10.1145/2556288.2557165> event-place: Toronto, Ontario, Canada.
- [22] Takato Horii, Yukie Nagai, and Minoru Asada. 2013. Touch and Emotion: Modeling of developmental differentiation of emotion lead by tactile dominance. In

- 2013 *IEEE Third Joint International Conference on Development and Learning and Epigenetic Robotics (ICDL)*. IEEE, 1–6.
- [23] Yingying Jiang, Wei Li, M. Shamim Hossain, Min Chen, Abdulhameed Alelaiwi, and Muneer Al-Hammadi. 2020. A snapshot research and implementation of multimodal information fusion for data-driven emotion recognition. *Information Fusion* 53 (Jan. 2020), 209–221. <https://doi.org/10.1016/j.inffus.2019.06.019>
- [24] Michael Karlesky and Katherine Isbister. 2014. Fidget widgets: designing for the physical margins of digital workspaces. In *Proceedings of the 8th International Conference on Tangible, Embedded and Embodied Interaction*. ACM, 301–304.
- [25] Michael Karlesky and Katherine Isbister. 2016. Understanding Fidget Widgets: Exploring the Design Space of Embodied Self-Regulation. In *Proceedings of the 9th Nordic Conference on Human-Computer Interaction*. ACM, 38.
- [26] Todd B Kashdan and Michael F Steger. 2006. Expanding the topography of social anxiety: An experience-sampling assessment of positive emotions, positive events, and emotion suppression. *Psychological Science* 17, 2 (2006), 120–128.
- [27] Michael J Kofler, Joseph S Raiker, Dustin E Sarver, Erica L Wells, and Elia F Soto. 2016. Is hyperactivity ubiquitous in ADHD or dependent on environmental demands? Evidence from meta-analysis. *Clinical psychology review* 46 (2016), 12–24.
- [28] Tara L Kraft and Sarah D Pressman. 2012. Grin and bear it: The influence of manipulated facial expression on the stress response. *Psychological science* 23, 11 (2012), 1372–1378.
- [29] Po-Ming Lee, Wei-Hsuan Tsui, and Tzu-Chien Hsiao. 2015. The Influence of Emotion on Keyboard Typing: An Experimental Study Using Auditory Stimuli. *PLOS ONE* 10, 6 (June 2015), e0129056. <https://doi.org/10.1371/journal.pone.0129056>
- [30] Shan Li and Weihong Deng. 2022. Deep Facial Expression Recognition: A Survey. *IEEE Transactions on Affective Computing* 13, 3 (July 2022), 1195–1215. <https://doi.org/10.1109/TAFFC.2020.2981446> arXiv:1804.08348 [cs].
- [31] Zachary C Lipton, John Berkowitz, and Charles Elkan. 2015. A critical review of recurrent neural networks for sequence learning. *arXiv preprint arXiv:1506.00019* (2015).
- [32] A. F. M. Nazmul Haque Nahin, Jawad Mohammad Alam, Hasan Mahmud, and Kamrul Hasan. 2014. Identifying emotion by keystroke dynamics and text pattern analysis. *Behaviour & Information Technology* 33, 9 (Sept. 2014), 987–996. <https://doi.org/10.1080/0144929X.2014.907343>
- [33] National Institutes of Health (NIH). 2023. Project Information - NIH Project Number 1R01NS109423-01. <https://reporter.nih.gov/project-details/10064501> Accessed: 2024-01-24.
- [34] Soujanya Poria, Erik Cambria, Rajiv Bajpai, and Amir Hussain. 2017. A review of affective computing: From unimodal analysis to multimodal fusion. *Information fusion* 37 (2017), 98–125.
- [35] Mark D Rapport, Jennifer Bolden, Michael J Kofler, Dustin E Sarver, Joseph S Raiker, and R Matt Alderson. 2009. Hyperactivity in boys with attention-deficit/hyperactivity disorder (ADHD): a ubiquitous core symptom or manifestation of working memory deficits? *Journal of abnormal child psychology* 37 (2009), 521–534.
- [36] Thomas L Rodebaugh, Carol M Woods, David M Thissen, Richard G Heimberg, Dianne L Chambless, and Ronald M Rapee. 2004. More information from fewer questions: the factor structure and item properties of the original and brief fear of negative evaluation scale. *Psychological assessment* 16, 2 (2004), 169.
- [37] Dustin E Sarver, Mark D Rapport, Michael J Kofler, Joseph S Raiker, and Lauren M Friedman. 2015. Hyperactivity in attention-deficit/hyperactivity disorder (ADHD): Impairing deficit or compensatory behavior? *Journal of abnormal child psychology* 43, 7 (2015), 1219–1232.
- [38] Daniel Shapiro, Zeping Zhan, Peter Cottrell, and Katherine Isbister. 2019. Translating affective touch into text. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [39] Thanyathorn Thanapatttheerakul, Katherine Mao, Jacqueline Amoranto, and Jonathan Chan. 2018. *Emotion in a Century: A Review of Emotion Recognition*. <https://doi.org/10.1145/3291280.3291788> Pages: 8.
- [40] George A. Tsihrintzis, Maria Virvou, Efthymios Alepis, and Ioanna-Ourania Stathopoulou. 2008. Towards Improving Visual-Facial Emotion Recognition through Use of Complementary Keyboard-Stroke Pattern Information. *Fifth International Conference on Information Technology: New Generations (itng 2008)* (2008). https://www.academia.edu/49511272/Towards_Improving_Visual_Facial_Emotion_Recognition_through_Use_of_Complementary_Keyboard_Stroke_Pattern_Information
- [41] Karen Vanderloock, Vero Vanden Abeele, Johan AK Suykens, and Luc Geurts. 2013. The skweezee system: enabling the design and the programming of squeeze interactions. In *Proceedings of the 26th annual ACM symposium on User interface software and technology*. ACM, 521–530.
- [42] Yan Wang, Wei Song, Wei Tao, Antonio Liotta, Dawei Yang, Xinlei Li, Shuyong Gao, Yixuan Sun, Weifeng Ge, Wei Zhang, and Wenqiang Zhang. 2022. A systematic review on affective computing: emotion models, databases, and recent advances. *Information Fusion* 83-84 (July 2022), 19–52. <https://doi.org/10.1016/j.inffus.2022.03.009>
- [43] Richard W Woodcock, Kevin S McGrew, Nancy Mather, and Fredrick A Schrank. 2003. Woodcock-Johnson III diagnostic supplement to the tests of cognitive abilities. *Itasca, IL: Riverside* 10 (2003), 003435520104400407.
- [44] Sydney Zentall. 1975. Optimal stimulation as theoretical basis of hyperactivity. *American Journal of Orthopsychiatry* 45, 4 (1975), 549.
- [45] Sydney S Zentall, Steven D Falkenberg, and Linda B Smith. 1985. Effects of color stimulation and information on the copying performance of attention-problem adolescents. *Journal of abnormal child psychology* 13, 4 (1985), 501–511.
- [46] Sydney S Zentall and Martha J Meyer. 1987. Self-regulation of stimulation for ADD-H children during reading and vigilance task performance. *Journal of Abnormal Child Psychology* 15 (1987), 519–536.
- [47] Zhihong Zeng, M. Pantic, G.I. Roisman, and T.S. Huang. 2009. A Survey of Affect Recognition Methods: Audio, Visual, and Spontaneous Expressions. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 31, 1 (Jan. 2009), 39–58. <https://doi.org/10.1109/TPAMI.2008.52>
- [48] Barbra Zupan and Michelle Eskritt. 2020. Eliciting emotion ratings for a set of film clips: A preliminary archive for research in emotion. *The Journal of Social Psychology* 160, 6 (2020), 768–789.

A SUPPLEMENTARY MATERIAL

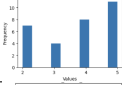
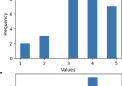
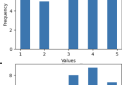
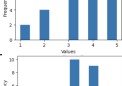
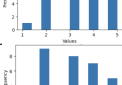
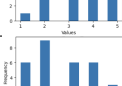
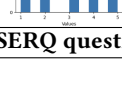
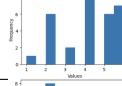
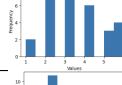
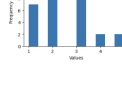
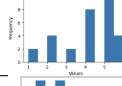
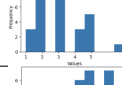
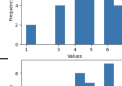
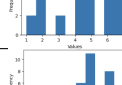
This study utilized standard psychological instruments to assess participants’ social anxiety and efforts at anxiety management in response to stress tests. Table 1 provide the self-assessment questionnaires and the distribution of participant responses to each question.

The SSA questionnaire (Table 1) contains seven questions designed to assess social anxiety levels by focusing on participants’ perceptions and reactions in social situations, especially regarding feelings of being evaluated or observed by others. Participants are asked to indicate the extent to which they experienced certain feelings during the interview task on a Likert scale, from 1 (‘not at all or very slightly’) to 5 (‘extremely’). The distribution of responses is different for each question, as shown by the histograms in column 2. None of these distributions are narrow about the median, making the median guess a suitable random baseline.

The third column of Table 1 provides the results obtained from our classification analysis of the associated fidgeting data, which treats each answer to a given SSA question as a categorical variable. This approach represents a somewhat difficult framing of the analysis task because it counts near misses in predicting a numerical value as misclassifications. Nevertheless, the results show accuracy significantly above the random baseline of 20% across all SSA questions, indicating that hand-fidgeting contains a strong signal regarding affective state.

The SERQ questionnaire (Table 1) contains eight questions designed to assess participants’ efforts at regulating their emotions. The questions posit strategies for mitigating or controlling negative emotions and ask participants to indicate the extent of their agreement on a Likert scale, here from 1 (‘strongly disagree’) to 7 (‘strongly agree’). As before, the distribution of responses (column 2) is different for each question, and none of these distributions are narrow about the median, making the median guess a suitable random baseline.

Table 1: Ground truth distribution of SSA and SERQ questions

Ground truth distribution of SSA questions and associated classification analysis		
SSA Questions	Answer Spread	Classification Acc (%)
Q1) I worried about what other people thought of me.		69.00
Q2) I was afraid other people would notice my shortcomings.		58.32
Q3) I was afraid that others did not approve of me.		52.32
Q4) I was worried that I would say or do the wrong things.		80.50
Q5) When I was talking to someone, I was worried about what they were thinking of me.		57.99
Q6) I felt uncomfortable and embarrassed when I was the center of attention.		81.11
Q7) I found it hard to interact with people.		69.65
Ground truth distribution of SERQ questions on a Likert scale		
SERQ Questions	Answer Spread	
Q1) When I wanted to feel more positive emotions, I changed what I was thinking about.		
Q2) I kept my emotions to myself.		
Q3) When I wanted to feel less negative emotion (such as sadness or guilt), I changed what I was thinking about.		
Q4) When I was feeling positive emotions, I was careful not to express them.		
Q5) I controlled my emotions by not expressing them.		
Q6) I controlled my emotions by changing the way I thought about the situation I was in.		
Q7) When I was feeling negative emotions, I made sure not to express them.		
Q8) When I wanted to feel less negative emotion, I changed the way I was thinking about the situation.	